

Getting Started / Setup Guide

Getting Started / Setup Guide - DRAFT

1. About the Valley Development Kit

The Alps Alpine Valley Development Kit provides both hardware and software to help users quickly connect, evaluate, and integrate a variety of Alps Alpine components into their current system or setup. The kit includes interface options that simplify component and device connections, while the graphical user interface allows users to instantly visualize component outputs in a clear and useful way.

This kit helps users determine whether a particular sensor, or a group of sensors, will meet their application needs while also supporting a smoother path from evaluation to integration.

What you can do with Valley

The Alps Alpine Valley Development Kit is a modular evaluation and prototyping platform for Alps Alpine sensor and HMI technologies. Valley works with the PEAK Development Suite to simplify device connection, configuration, and live data visualization.

- Connect compatible Alps Alpine sensor and HMI boards using Qwiic I²C and other supported interfaces.
- Use PEAK to discover connected peripherals and visualize live sensor and HMI data.
- Evaluate multiple supported sensors within a single development setup.
- Use configurable GPIO, buttons, and LEDs for development and prototyping.
- Evaluate tactile-feedback applications using Valley's dual haptic drivers.
- Create haptic waveforms, rules, and actions using supported PEAK functionality.

2. Before You Begin

What comes in the kit

- **Valley Development Kit** - Part Number XY00053A01.
 - **Valley EVK**
 - **Barrel Jack Power Supply (12v/ 2A)**
 - **USB micro to USB A cable.**
- **Peripheral Device(s) -depending on kit configuration**
 - **HMI Boards**
 - **Sensor Boards**
 - **Cirque Touch Boards**

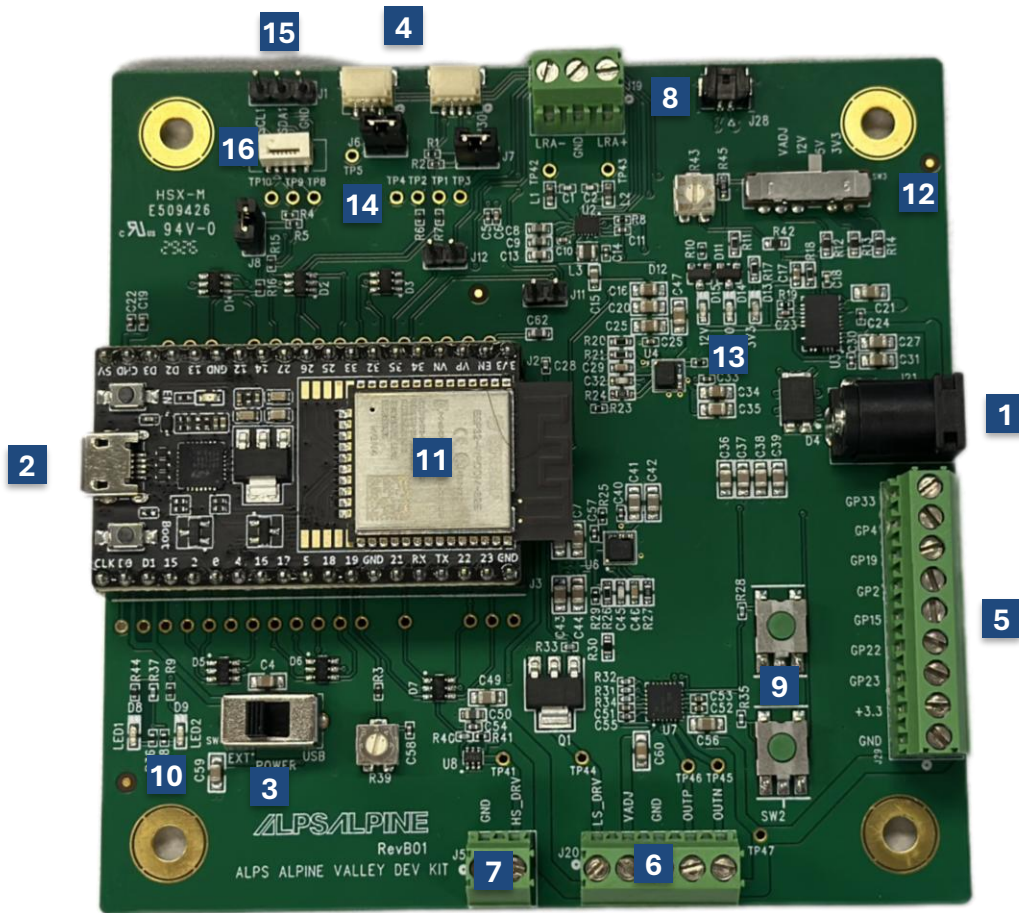
What else is needed

- **Windows PC (Windows 10 or higher)**
- **PEAK Application**
<https://github.com/Alps-Alpine-North-America-Inc/Alps-Alpine-Peak-Development-Suite-PC-Application>

This guide walks you through setting up the Valley Development Kit, connecting supported peripheral boards, and using the PEAK Development Suite. It also introduces the available hardware configurations and provides the steps needed to quickly begin evaluating Alps Alpine devices.

Getting started goal: Connect one ThumbPointer board, open PEAK, move the joystick, and confirm the live X/Y data changes. Press the center switch and confirm the button state changes.

3. Hardware Overview



Callout	Interface / Feature	Reference Designator	Purpose
1	Barrel-Jack Power Input	J21	Connect the supplied 12 V / 2 A external power supply . For normal operation, connect external power before connecting USB.
2	Micro-USB Connector	N/A	Connects the Valley board's ESP32 to a Windows PC running the PEAK Development Suite . Use the supplied Micro-USB to USB-A cable.
3	Power Source Selector	SW4	Selects between EXT external power and USB power. Some Valley functions and drivers are unavailable when operating from USB power only. See below section for details.
4	Qwiic Connectors	J22 & J30	Connect compatible Alps Alpine HMI and sensor peripheral boards over the Qwiic / I ² C0 interface.
5	9-pin GPIO Terminal	J29	Provides access to configurable ESP32 GPIO for external inputs, outputs, and custom development.

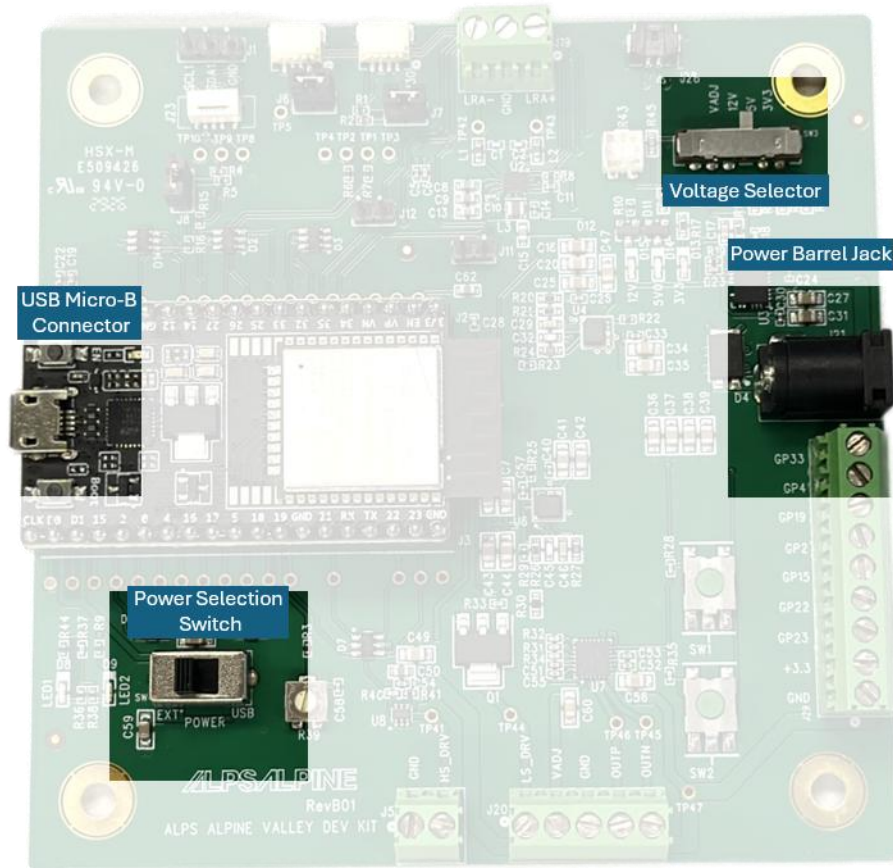
Callout	Interface / Feature	Reference Designator	Purpose
6	5-Pin Driver Terminal	J20	Provides access to the Valley H-Bridge (Differential) Driver and Low-Side Driver outputs for haptic evaluation and other supported external loads. See Section 6 for electrical ratings and connection details.
7	2-Pin High-Side Driver Terminal	J5	Provides a switched high-side output for driving supported external loads using the selected Valley driver supply voltage. See Section 6 for electrical ratings and connection details.
8	Audio Driver Connectors	J19 & J28	Provides dedicated connections to the Valley Audio Driver for supported haptic actuators. The included haptic actuator can be connected directly, or a compatible actuator can be connected using the screw terminal.
9	Tact Switches	SW1 & SW2	User-configurable digital inputs for evaluation, development, and PEAK interaction.
10	LEDs	LED2	User-configurable visual indicators for evaluation and development.
11	ESP32-DEVKITC-32E	J2 & J3	Main controller for the Valley Development Board. Runs the Valley firmware, communicates with PEAK over USB, and controls the board's configurable I/O and peripheral interfaces.
12	Voltage Selector Slide Switch	SW3	Selects the supply voltage used by supported Valley driver outputs. Available settings include 3.3 V, 5 V, 12 V, and Adjustable. See Section 3.1 for additional details and limitations.
13	Power LEDs	D13-D15	Provide visual indication of the selected/active voltage configuration. See Section 3.1 for voltage-selection behavior.
14	Configurable Jumpers	J6-J8, J11, J12	Valley Development Board includes five removable configuration jumpers. See more information below.
15	I2C1 Pin Header	J1	Connection to the I2C 1 channel. Contains SCL, SDA and GND.
16	FFC Header	J23	Ribbon cable connection compatible Cirque development boards over the I ² C1 interface.

3.1 On Board configurations

• Power Configuration

The Valley Development Board supports multiple power configurations.

For normal operation, use the external power configuration shown below with the 12 V barrel-jack power supply included with the kit. This configuration provides the power required for the board's drivers and on-board components to operate as intended.



For applications that primarily use the on-board **ESP32** and do not require all Valley peripherals or drivers, the board can also be powered through USB only.

To use USB power, move the power selector switch from **EXT** to **USB**. You may then remove the barrel jack connection.

Note: USB power is intended for limited-use configurations. Use the external 12 V supply for normal Valley operation and when using the board's full set of peripherals and drivers.

• Voltage Selector

The Valley Development Board includes a **voltage selector** switch that controls the supply voltage used by several on-board driver circuits. The available selections are:

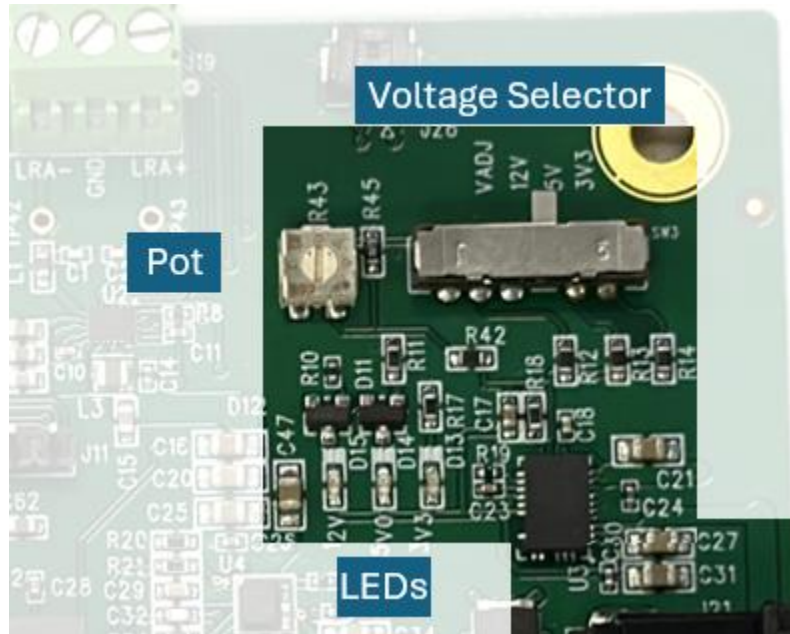
- **3.3 V**
- **5 V**
- **12 V**
- **Adjustable** — set using potentiometer **R43**

When **Adjustable** is selected, R43 is used to set the output voltage between 0 V and 12 V. The corresponding indicator LED illuminates to show the currently selected voltage setting.

The voltage selector controls the supply voltage for the following functions:

- High-Side Driver
- Differential Driver
- Voltage output on J20, Pin 2

Voltage Selector Note: The differential driver has a minimum input voltage of 4.5V. If you have the voltage selector set to 3.3V, it will not operate.



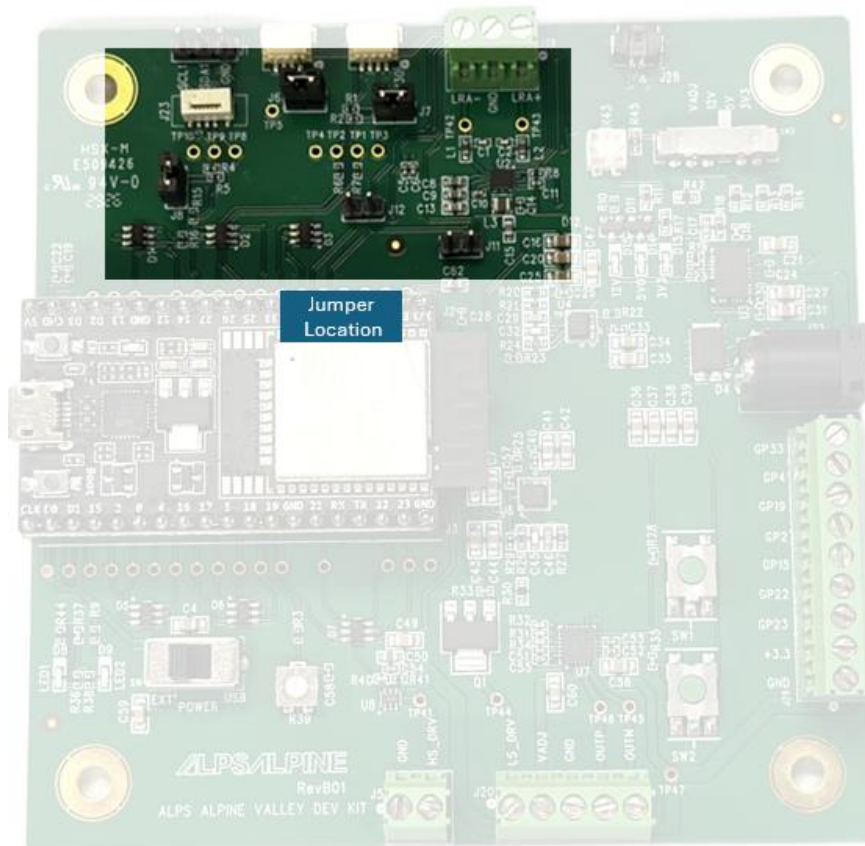
• Jumper Configuration

The Valley Development Board includes five removable configuration jumpers. The table below describes the function of each jumper and its default configuration.

Ref Des	Function	Default Position
J8	Enables the I ² C1 pull-up resistors. Remove this jumper when external pull-up resistors are already provided on the bus or when troubleshooting I ² C timing issues with multiple connected peripherals.	Populated
J6	Enables power sourcing on J22. Remove this jumper if the Qwiic bus is already being powered by an external device to prevent multiple devices from sourcing power onto the same bus.	Populated
J7	Enables the I ² C0 pull-up resistors. Remove this jumper when external pull-up resistors are already provided on the bus or when	Populated

	troubleshooting I ² C timing issues with multiple connected peripherals.	
J12	Connects TRIG3 for the audio driver. This connection is not required for normal audio-driver operation and is intended for advanced use. When J12 is populated, GPIO 4 is no longer available for general-purpose I/O.	Unpopulated
J11	Connects TRIG2 for the audio driver. This connection is not required for normal audio-driver operation and is intended for advanced use. When J11 is populated, GPIO 32 is no longer available for general-purpose I/O.	Unpopulated

Recommended Configuration: For normal use, leave all jumpers in their default positions.



- **0 Ω Configuration Resistors**

The Valley Development Board also includes several **0 Ω configuration resistors** that connect ESP32 GPIO signals to on-board functions.

These resistors may be removed to make additional ESP32 I/O available for custom applications. **Removal is not recommended for normal Valley operation**, as doing so will disconnect the associated on-board function.

Ref Des	Function	I/O Function
R9	LED2	I/O_4
R44	LED1	I/O_0
R3	POT_EN	IN_39
R7	I ² C0 SCL	I/O_26
R6	I ² C0 SDA	I/O_27
R15	I ² C1 SDA	I/O_14
R16	I ² C1 SCL	I/O_13

Advanced Use Only: Modifying the 0 Ω configuration resistors changes the default Valley hardware configuration and should only be performed when developing a custom application that requires the associated ESP32 I/O.

4. Installing the PEAK Development Suite

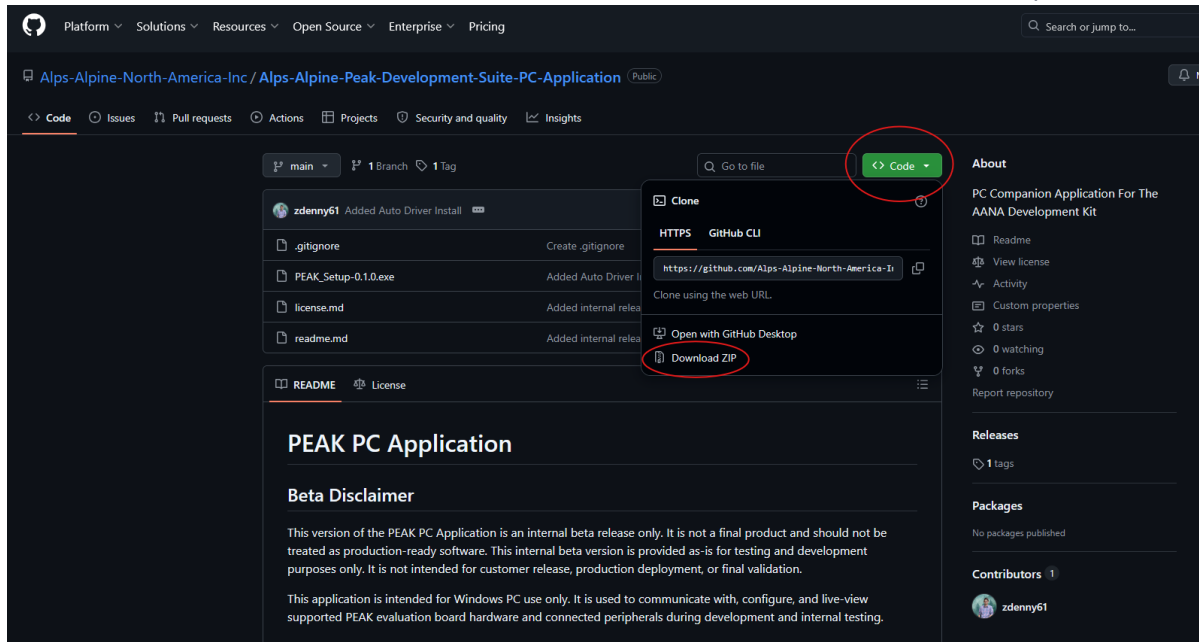
PEAK is the Windows companion application used to communicate with Valley, configure supported peripherals, and view live peripheral data.

Check for updates: Before starting, check the Alps Alpine GitHub repository and download the latest stable, customer-approved PEAK release. Do not rely on an older installer saved locally.

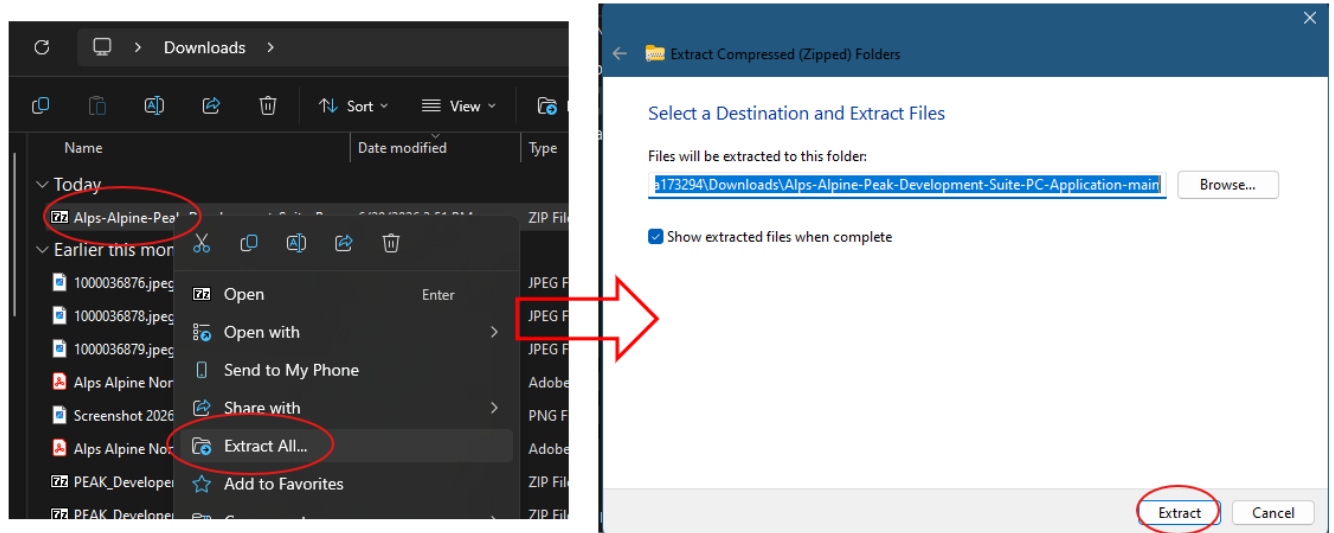
You can also opt for automatic updates in the PEAK application when the installation is completed.

Step 1 - Download the latest stable PEAK release

1. Open the PEAK GitHub repository from the Valley product page or use this [link](#).
2. Download the latest stable Windows release approved for customer use.

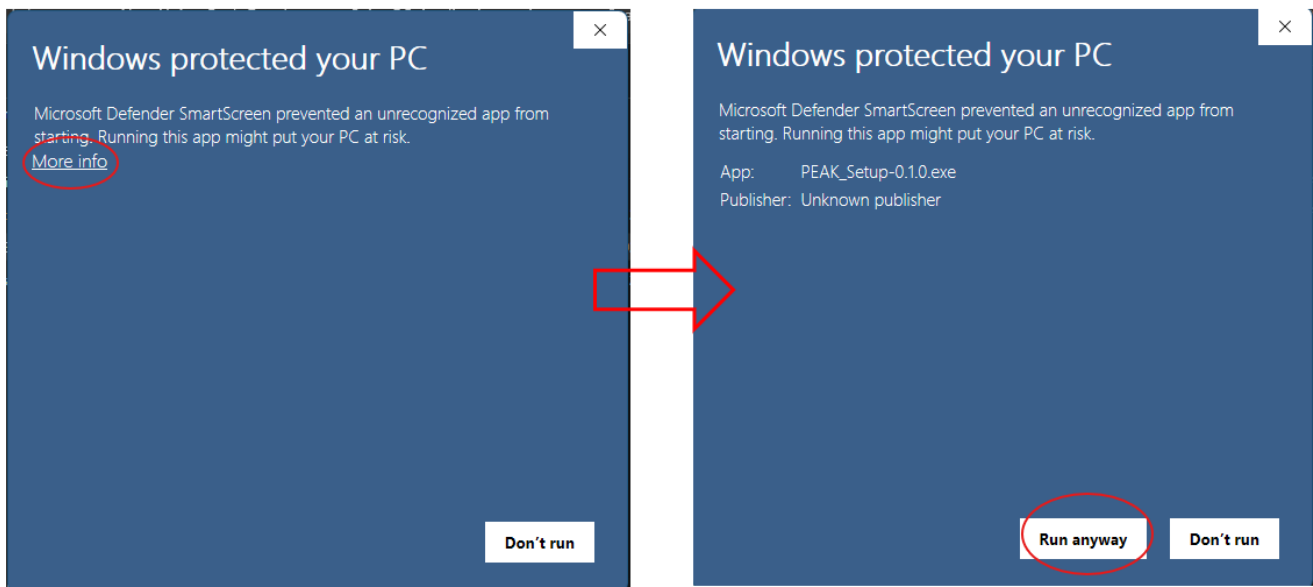


3. Extract the downloaded ZIP package if the release is provided as a compressed archive.

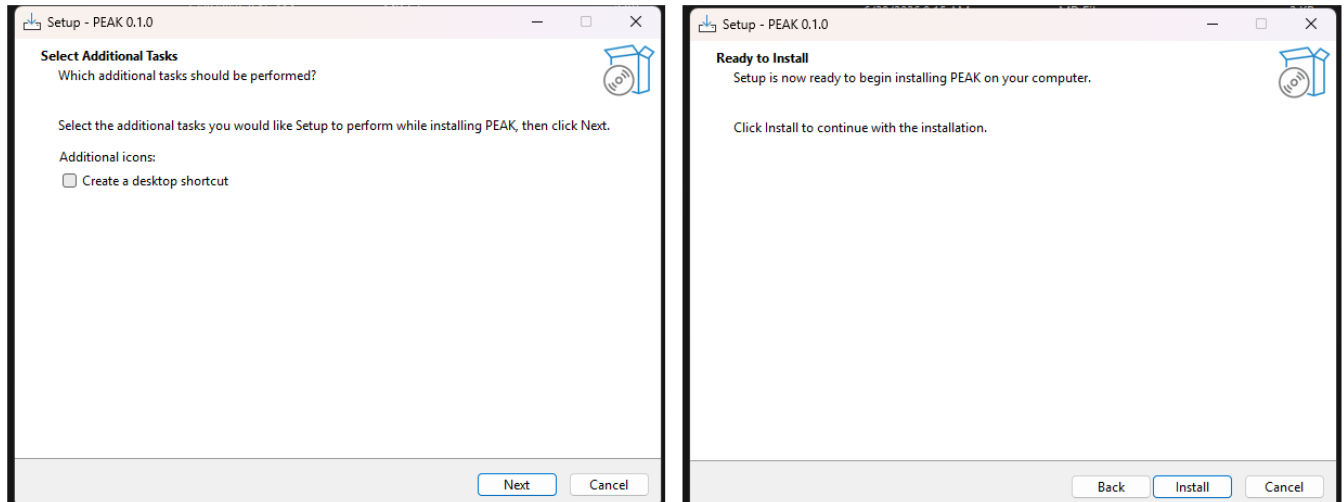


Step 2 - Run the PEAK installer

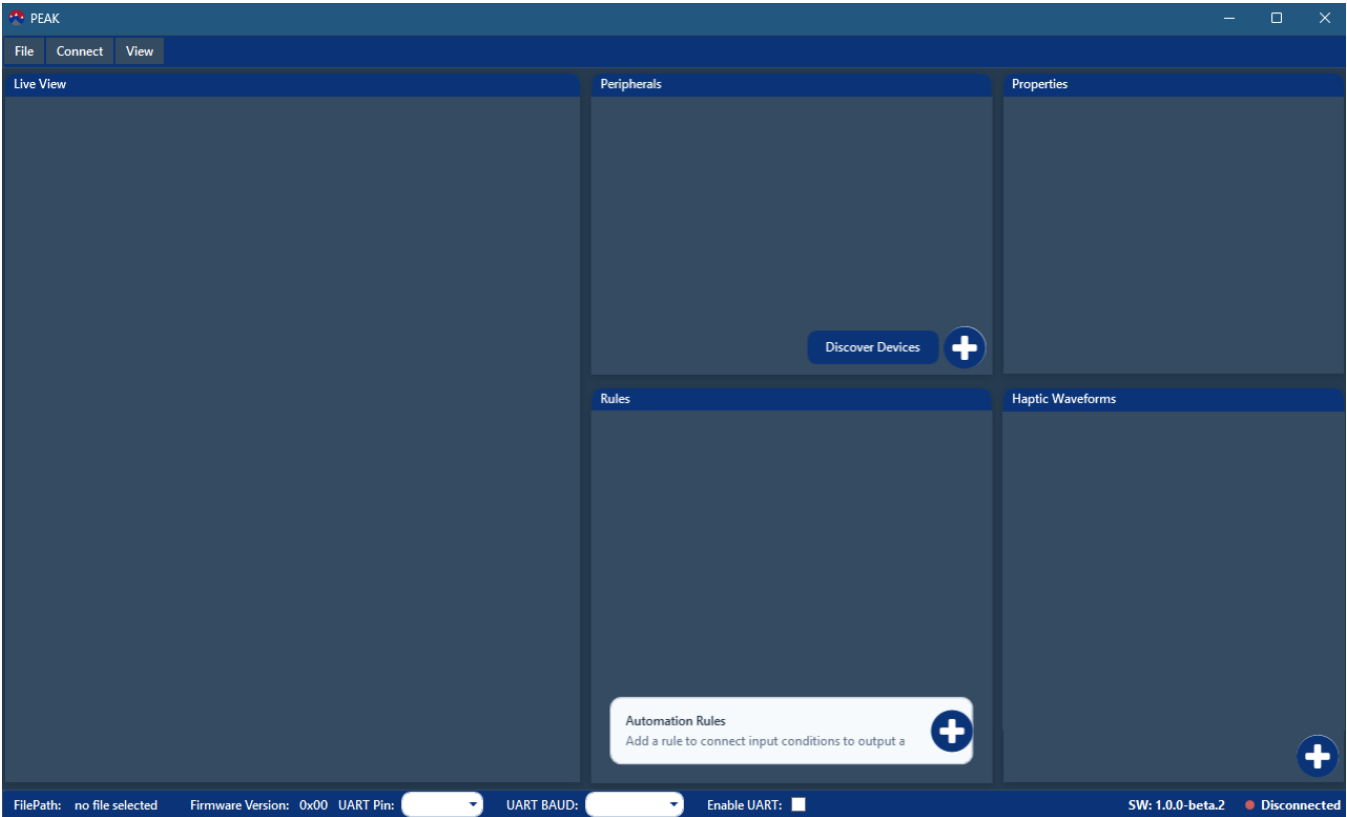
1. Open the extracted PEAK installer.
2. If Microsoft Defender SmartScreen appears, first confirm the installer was downloaded from the official Alps Alpine GitHub repository. Select More info, then Run anyway to continue.



3. Follow the PEAK Setup Wizard and complete the installation.



4. Launch PEAK after installation finishes.



5. Getting started demo – ThumbPointer Board

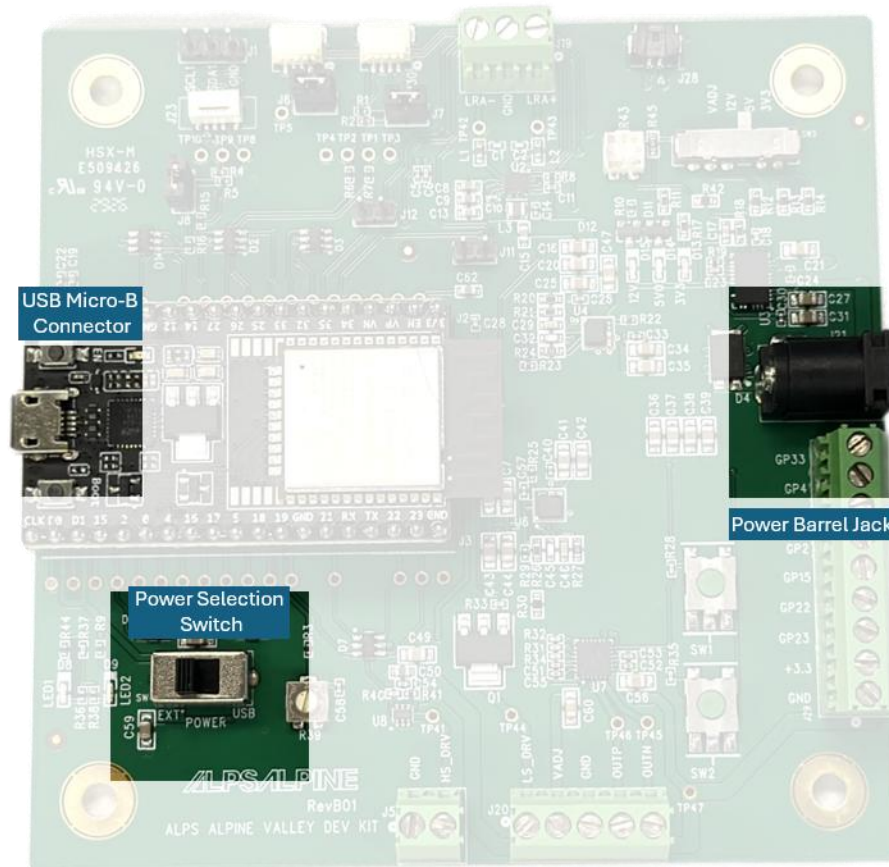
This section walks through a basic ThumbPointer joystick demonstration to verify that the Valley Development Kit, PEAK Development Suite, and connected peripheral are operating correctly. Other peripherals will be set up similarly.

Step 1 - Connect the barrel-jack power supply

1. Connect the approved external power supply to the Valley barrel-jack input first.

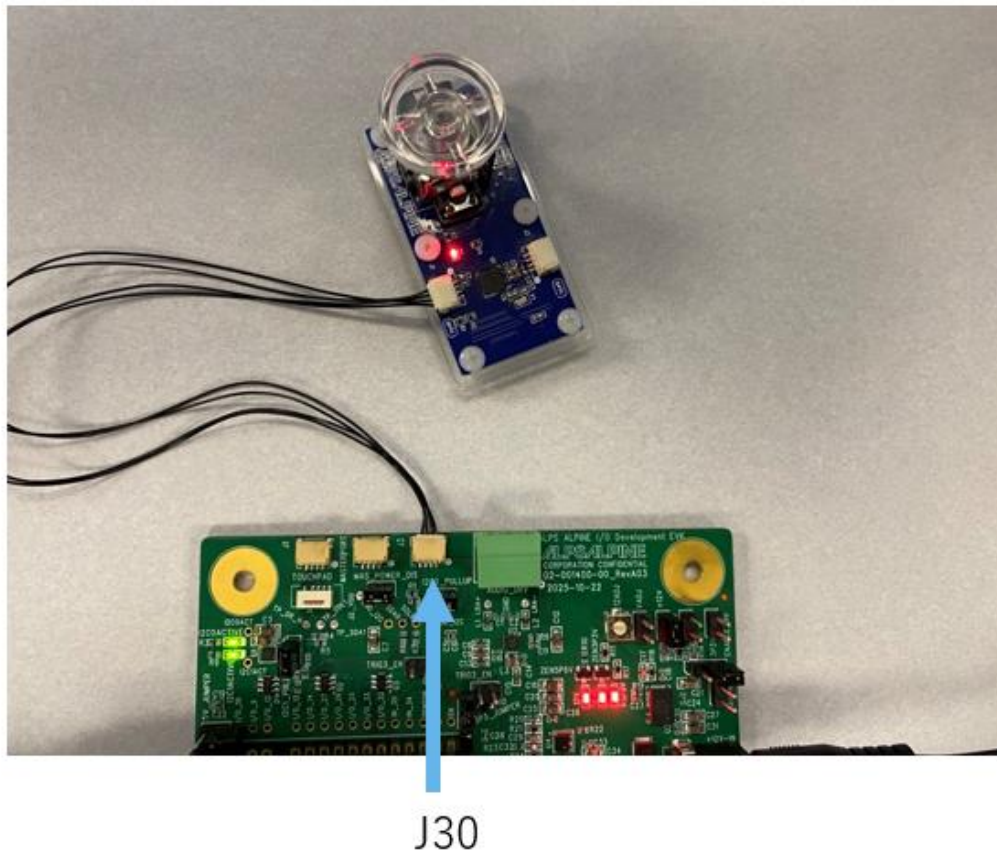
Step 2 - Connect USB to the PC

1. After barrel-jack power is connected, connect Valley to the Windows PC using the supported USB cable.
2. Ensure the power selection switch is set to EXT.



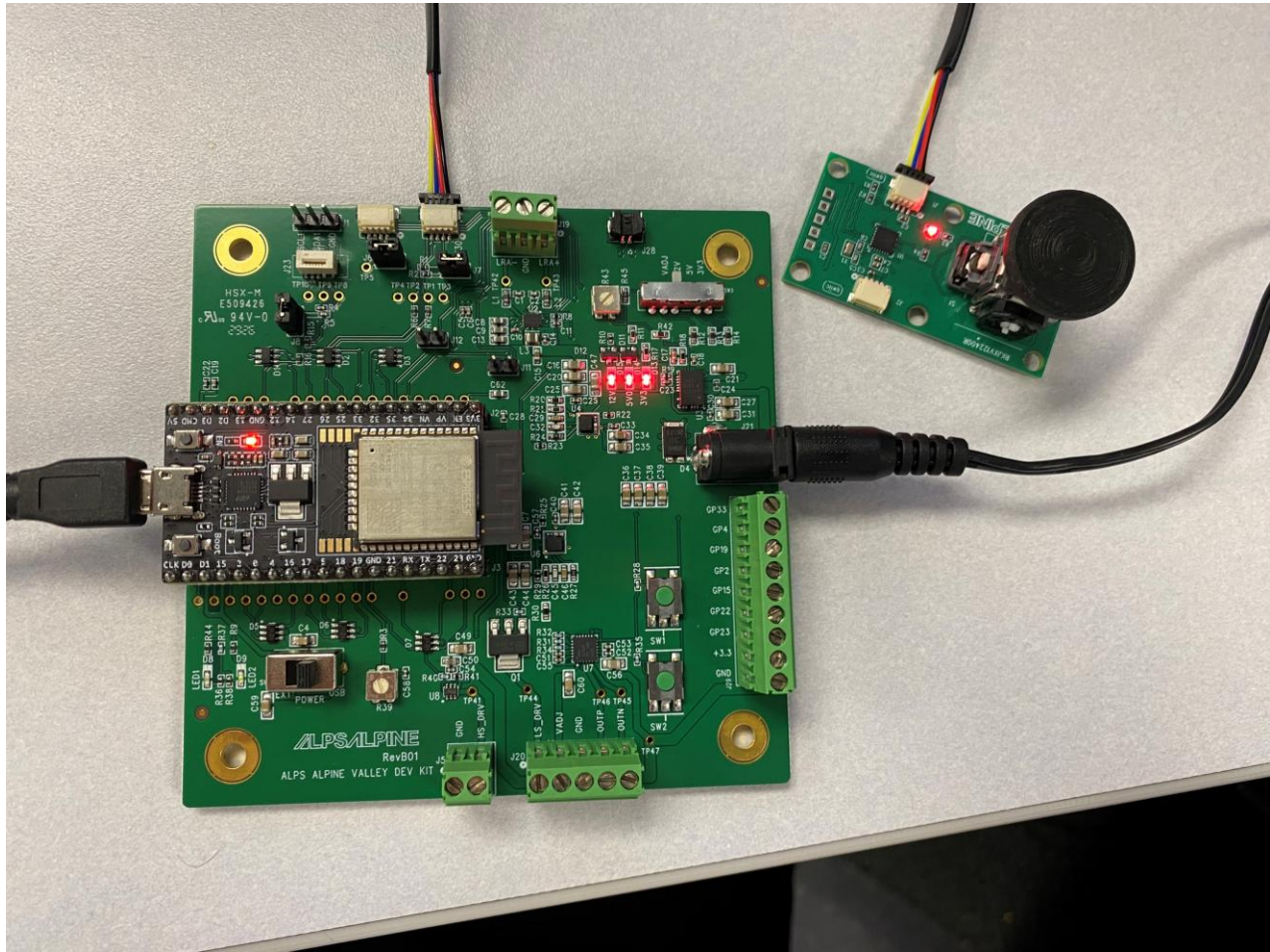
Step 3 - Connect the ThumbPointer board

1. Connect one end of a Qwiic cable to either Valley Qwiic I2C host port.
2. Connect the other end of the Qwiic cable to the ThumbPointer HMI Board (Qwiic).
3. Place the board where the joystick can be moved freely during the demo.



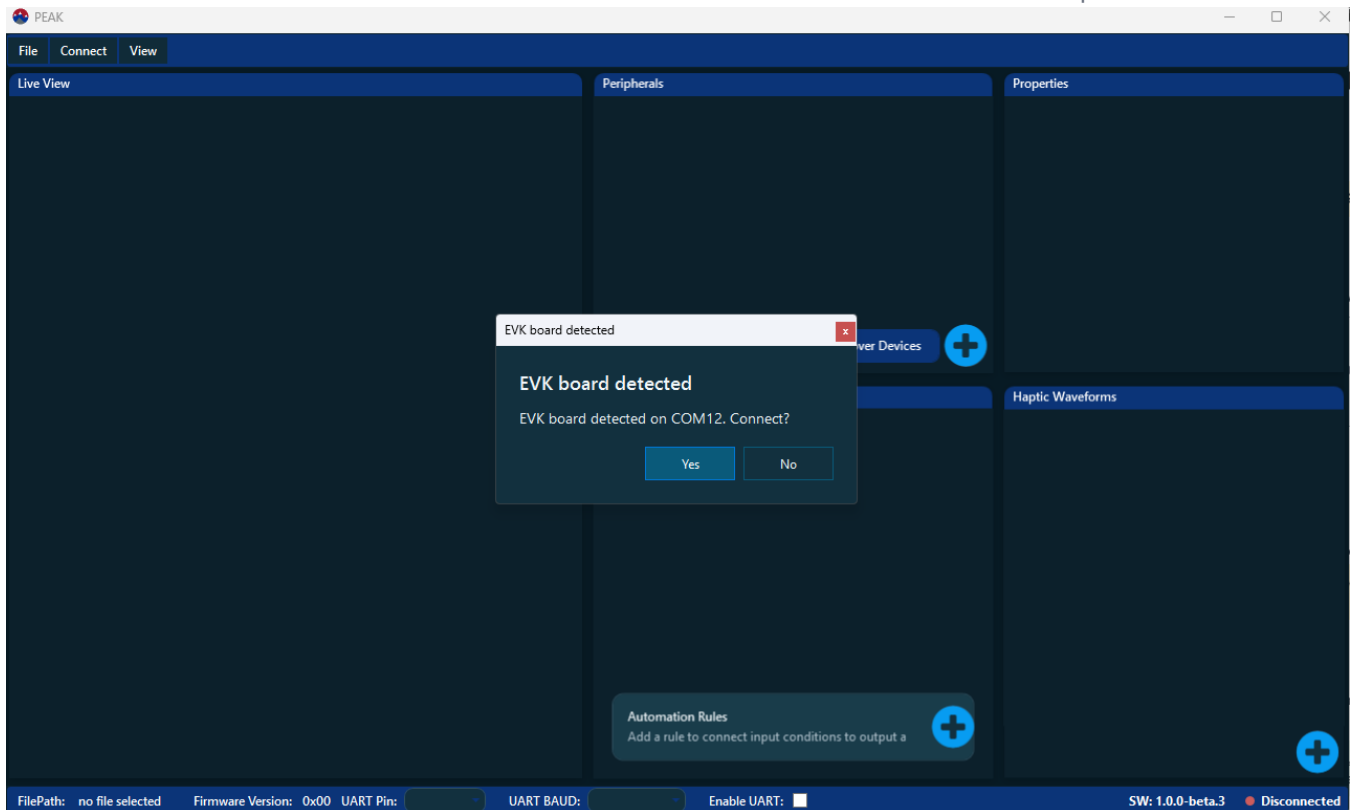
Step 4 - Verify power and connection

- Confirm LEDs are on for the sensor board and Valley EVK. The number of LEDs that illuminate on the Valley board depends on the voltage selector setting.
- Wait for Windows to finish detecting the device.
- If applicable, verify that a COM port appears in Windows Device Manager.



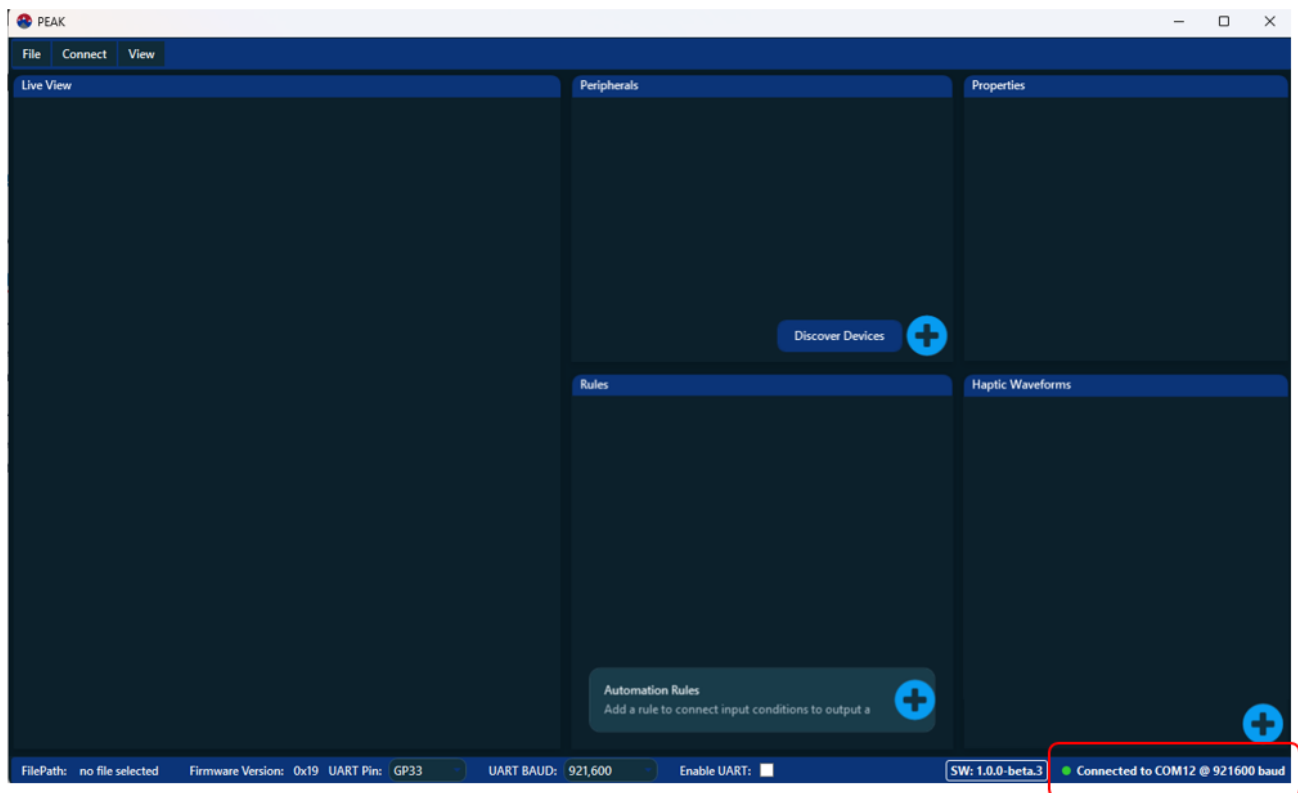
Step 5 - Open PEAK with the connected to Valley

1. Launch PEAK after Valley is powered and connected to the PC.
2. If PEAK displays an EVK board detected on COM# prompt, select Yes to connect.



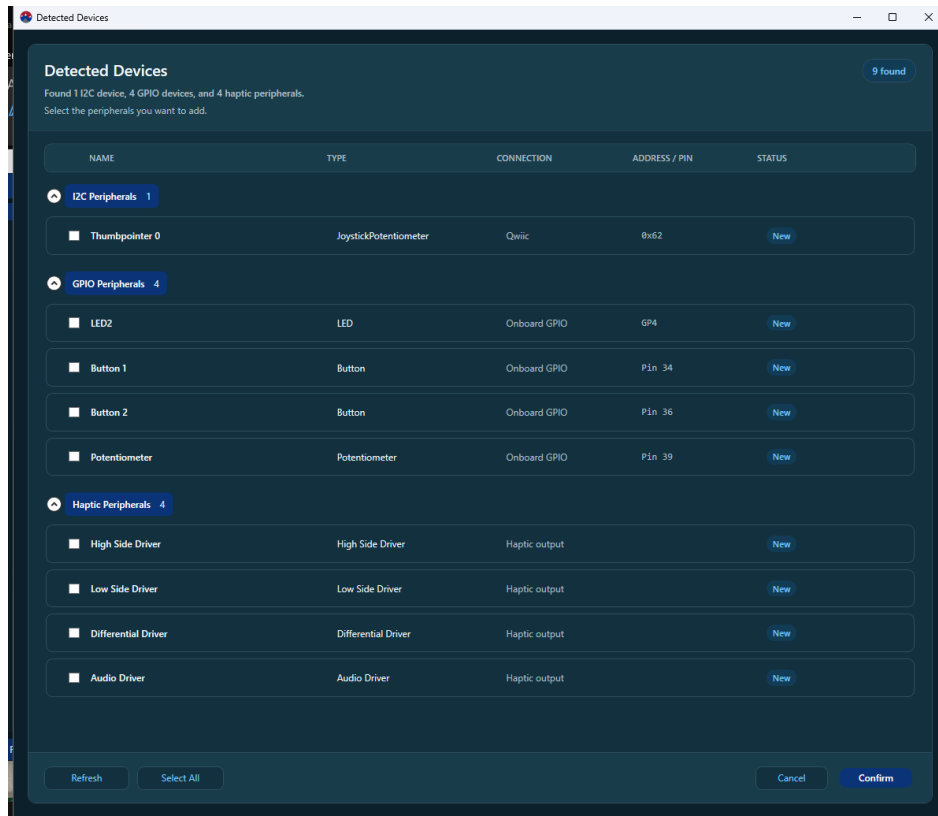
3. If the automatic prompt does not appear, use the Connect menu and select the COM port assigned to Valley.

4. Confirm PEAK shows Valley as connected before continuing.



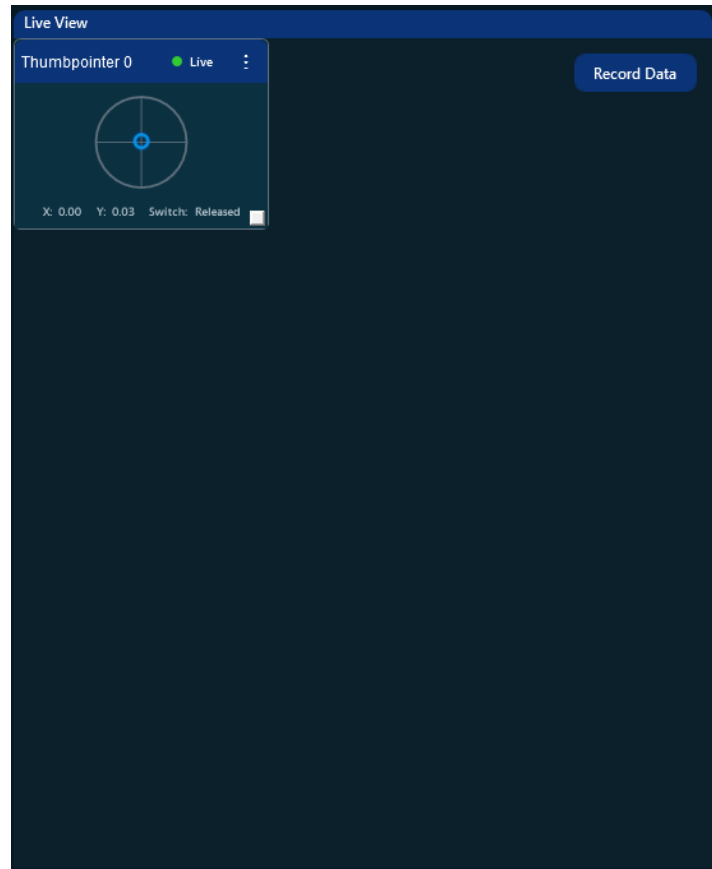
Step 6 - Discover the ThumbPointer peripheral

1. Select Discover Devices in the Peripherals area.
2. In the Detected Devices window, it contains all the devices the board detects whether its off board or on board.
Locate the connected ThumbPointer / joystick peripheral.



What PEAK may find: PEAK can discover I2C and onboard GPIO devices. For the first demo, only the ThumbPointer needs to be added to the Live View unless you intentionally want to demonstrate additional functions.

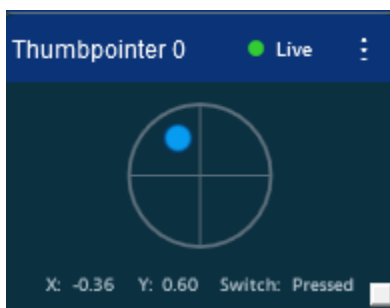
3. Select the ThumbPointer and choose Confirm.
4. Confirm the ThumbPointer is added to the Peripherals list and a corresponding element appears in the Live View.



Step 7 - Move the ThumbPointer and verify live data

1. Move the ThumbPointer joystick left and right and confirm the X-axis live data changes.
2. Move the joystick up and down and confirm the Y-axis live data changes.
3. Press the center of the ThumbPointer and confirm the push-button state changes in PEAK.

Expected result: The ThumbPointer data should respond immediately in the Live View as the joystick moves, and the center-push state should change when the joystick is pressed.

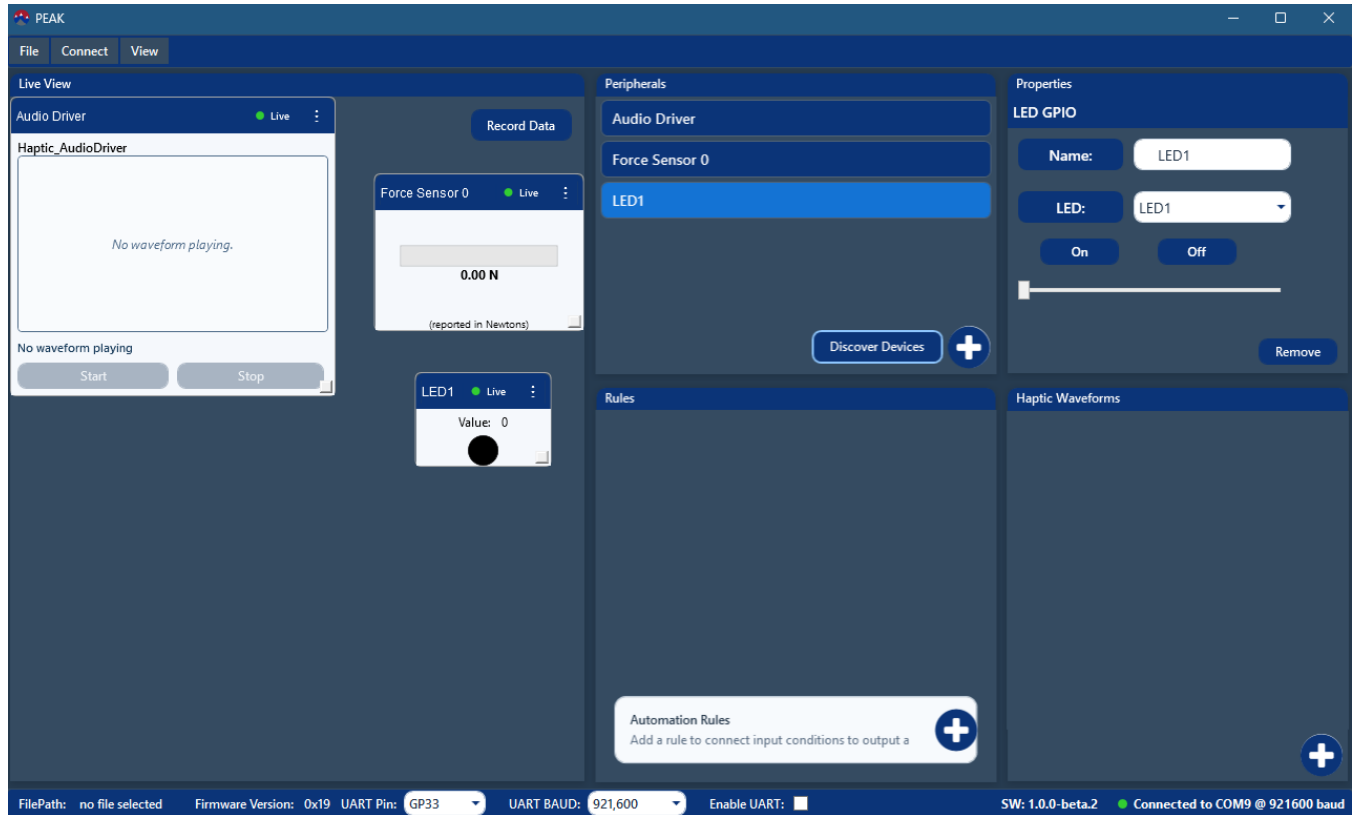


Additional supported HMI and sensor boards can be connected using the same general process. Multiple devices can be added to PEAK simultaneously. As the number of devices on a single I²C bus increases, polling latency may also increase.

6. Additional Valley & PEAK Capabilities

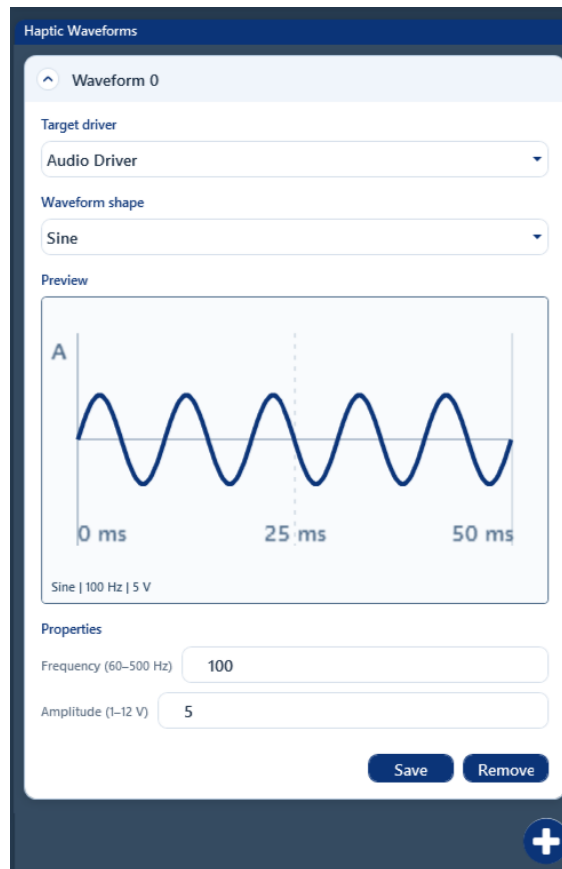
Live View

PEAK displays connected peripheral data in the Live View. Multiple device widgets can be arranged to observe sensor inputs and board outputs at the same time. You can record the data to output to a csv file.



Haptic waveforms

- Create a new waveform.
- Select the target haptic driver.
 - Audio Driver (Recommendation)
 - Differential Driver (H Bridge)
 - High Side driver
 - Low Side Driver
- Select a waveform shape.
- Enter supported waveform properties such as frequency and amplitude.
- Save the waveform for use by the haptic driver or an automation rule.



Note: The Audio Driver amplitude setting is independent of the Valley Voltage Selector because the audio driver uses an internal boost circuit. Always observe the electrical limits of the audio driver and connected actuator.



Automation rules

- Create a rule that links an input peripheral to an output peripheral.
- Define the input condition and threshold/value when applicable.
- Choose the output action, such as starting a saved haptic waveform.
- Save the rule and verify the input-to-output interaction.

Rules

▲

RULE

Input to output automation

Rule 1

Input

PERIPHERAL

Force Sensor 0

CONDITION

Value above

VALUE (N)

1

+

Output

PERIPHERAL

Audio Driver

ACTION

Start waveform

ACTION DETAILS

Waveform 0

+

Remove Rule

Save Rules

+

Discover Devices

- The PEAK application will automatically discover devices that are currently connected to the Valley board.

Detected Devices

Detected Devices

Found 1 I2C device, 4 GPIO devices, and 4 haptic peripherals.
Select the peripherals you want to add.

9 found

NAME	TYPE	CONNECTION	ADDRESS / PIN	STATUS
I2C Peripherals 1				
Thumbpointer 0	JoystickPotentiometer	Qwiic	0x62	New
GPIO Peripherals 4				
LED2	LED	Onboard GPIO	GP4	New
Button 1	Button	Onboard GPIO	Pin 34	New
Button 2	Button	Onboard GPIO	Pin 36	New
Potentiometer	Potentiometer	Onboard GPIO	Pin 39	New
Haptic Peripherals 4				
High Side Driver	High Side Driver	Haptic output		New
Low Side Driver	Low Side Driver	Haptic output		New
Differential Driver	Differential Driver	Haptic output		New
Audio Driver	Audio Driver	Haptic output		New

Refresh

Select All

Cancel

Confirm

Additional Valley hardware functions

- **Tact Switches**

Each tact switch has a 10k pull up.

- SW1 -> GPIO34
- SW2 -> GPIO36



- **9 Pin GPIO Terminal (J29)**



Each input has ESD protection up to +/- 8kV contact discharge. They connect directly to the ESP32 development board which operates at 3.3V logic level.

Warning: GPIO pins are **not 5V tolerant**. Connecting 5V logic directly to a GPIO pin will damage the ESP32.

+3.3 pin can be used as a 3.3V output to power external circuitry. When the power selection switch is set to EXT, it can source 5A.

When the power switch is set to USB, it is tied to 3.3V output regulator on the ESP32 dev board and is limited to 500mA.

- **5 Pin Driver Terminal**



The 5-pin driver terminal provides access to the Valley board's Differential (H-Bridge) Driver and Low-Side Driver outputs.

- **Differential Driver (H – Bridge)**

The Differential Driver is implemented using the Texas Instruments DRV8256PRGER H-Bridge Motor Driver. It can be used to drive supported haptic actuators as well as other loads requiring bidirectional control, such as a brushed DC motor. The DRV8256 includes integrated current regulation and protection features intended for driving inductive loads.

On the Valley board, the H-Bridge supply voltage is controlled by the Voltage Selector switch. Therefore, the available driver output voltage depends on the selected voltage setting. The DRV8256 requires a minimum operating supply of approximately 4.5 V, so the Differential Driver is not available when the Valley voltage selector is set to 3.3 V.

Warning: For the Valley Development Kit, the recommended maximum output current is 2 A at 12 V. Do not exceed the electrical limits specified for the Valley board when connecting external loads.

- **Low Side Driver**

The Valley Development Board includes a Low-Side Driver implemented with the onsemi NCV8405ASTT1G self-protected low-side switch. The device is designed to switch the ground side of an external load and includes integrated current limiting and thermal protection.

Unlike the H-Bridge output, the low-side driver does not provide the load supply voltage. An external voltage source can be used to power the load, while the Valley low-side driver switches the return path to ground.

The external supply ground and Valley ground must be connected so that both systems share a common reference.

Low-side output is limited to 3 A maximum based on the board layout and the SOT-223 package implementation. Do not exceed the electrical limits specified for the Valley board.

- **2 Pin Driver Terminal**



The Valley Development Board includes a High-Side Driver implemented using the Vishay Si1865DL load switch with integrated level shifting. The high-side driver provides a switched positive supply to an external load and can be controlled through Valley/PEAK

Unlike the Low-Side Driver, an external load supply is not normally required because the High-Side Driver sources power from the Valley board. The output supply is controlled by the Valley Voltage Selector, allowing the selected board supply to be switched to the connected load.

The High-Side Driver can be used for supporting haptic actuators or other low-current loads that require a switched positive supply.

The high side driver is limited to 1.1A when the voltage selector switch is set to 12V.

7. Quick Troubleshooting

Problem	What to Check First	Next Step
Valley does not power on	Barrel-jack supply, connector seating, approved power source, and connection order.	Disconnect USB, verify barrel-jack power first, then reconnect USB.
Valley does not appear in PEAK	Windows device detection and correct COM port.	Reconnect USB; check Device Manager; relaunch PEAK.
ThumbPointer is not detected	Qwiic cable, ThumbPointer connection, I2C bus/address, and device discovery.	Reconnect the single ThumbPointer board and run device discovery again.
ThumbPointer appears but live data does not change	Move the joystick through X/Y travel, press the center switch, and check poll rate/settings.	Reconnect the Qwiic cable and verify the ThumbPointer device properties.